20 The table shows some values of x and y that satisfy the equation $y = a\cos x° + b$

x	0	30	60	90	120	150	180
y	3	$1 + \sqrt{3}$	2	1	0	$1 - \sqrt{3}$	-1

Find the value of y when $x = 45$

cos 0° = 1 and cos 90° = 0.

1 mark
Use exact trigonometric values

At x = 90: 1 = a × 0 + b, so b = 1.

1 mark
Find b from a suitable row

At x = 0: 3 = a × 1 + 1, so a = 2.

1 mark
Find both constants

$$\cos 45° = \frac{\sqrt{2}}{2}$$

$$y = 2 \times \frac{\sqrt{2}}{2} + 1 = 1 + \sqrt{2}$$

1 mark
Correct exact value

1 + √2

(Total for Question 20 is 4 marks)

21 Show that $\dfrac{6 - \sqrt{8}}{\sqrt{2} - 1}$ can be written in the form $a + b\sqrt{2}$ where a and b are integers.

(Total for Question 21 is 3 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA